

Segmentation and Subnetting

- ! • **Segmentation** means splitting a **network** into smaller parts.
 - Each part (or segment) works independently and has its own broadcast domain.
 - Traffic from one segment is separated from the others.
- **Benefits of segmentation:**
 - Increases **security**
 - Boosts **performance**
 - Makes **troubleshooting** easier

How a Computer Uses a Subnet Mask

- An **IPv4 address** has two parts:
 - The **Network ID** and the **Host ID**.
- A **subnet mask** tells devices which part of the IP is for the **network**, and which is for the **host**. !!!!
 - The number of **1s** in the subnet mask shows how many bits belong to the **network part**.
- **Example:**
 - IP: 192.168.123.132 → Binary: 11000000.10101000.01111011.10000100
 - Subnet: 255.255.255.0 → Binary: 11111111.11111111.11111111.00000000

Table 10-2 Default IPv4 subnet masks

Network class	Default subnet mask (binary)	Number of bits used for network information	Default subnet mask (dotted decimal)
A	11111111 00000000 00000000 00000000	8	255.0.0.0
B	11111111 11111111 00000000 00000000	16	255.255.0.0
C	11111111 11111111 11111111 00000000	24	255.255.255.0

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CIDR (Classless Interdomain Routing)

- **CIDR** helps organize IP addresses without using fixed classes.
- It uses a **slash (/)** to show how many bits are for the **network ID**.
- Example: 192.168.89.127/24
 - The **/24** means 24 bits are used for the **network part**.
 - This format is called a **CIDR block**.

Why Subnets?

- ! • **Example:** A company grows from 30 to hundreds of computers across 3 floors.
 - Only one **LAN** or **broadcast domain** exists.
 - A single **router** manages all traffic.
- **Solution:**
 - **Divide the network** using subnets.
 - Each floor gets its own **LAN** to improve traffic control and performance.

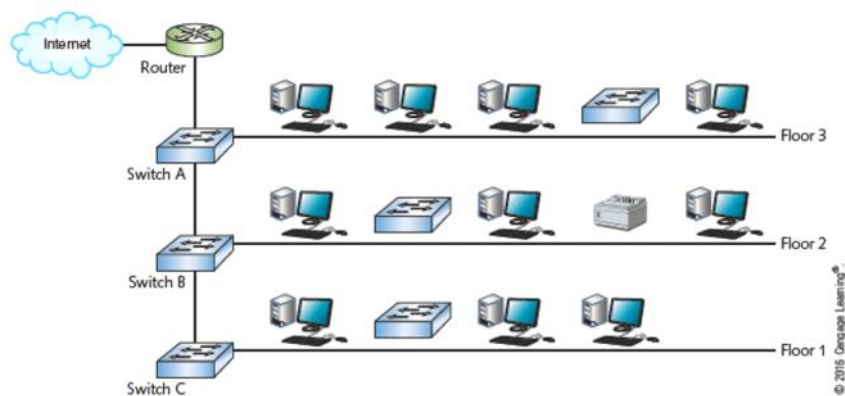


Figure 10-1 A single LAN with several switches and a router

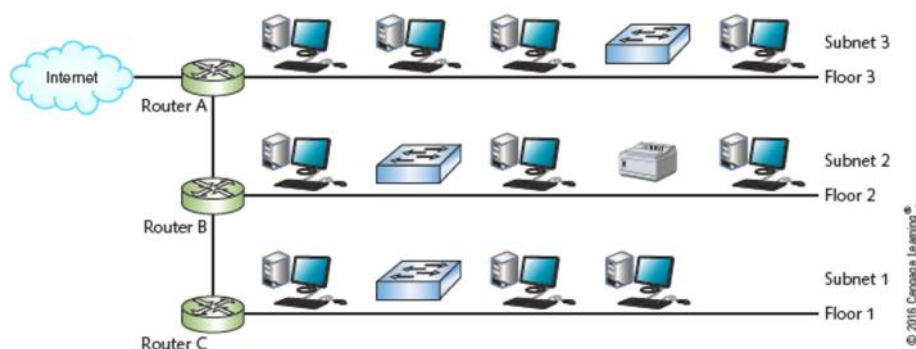


Figure 10-2 A separate subnet for each floor

Figure 10-1 – One LAN for All Floors

This diagram shows a single LAN shared across three floors.

- All devices are connected through **switches (A, B, C)**.
- Only **one router** connects entire network to the Internet.
- **No segmentation**, so all floors are part of the same broadcast domain.
- ✓ This setup causes **more traffic** and is **harder to manage**.

Figure 10-2 – One Subnet per Floor

This diagram shows each floor using a **separate subnet**.

- Each floor has its own **router** (Router A, B, C).
- Every subnet manages its own traffic.
- Devices on one floor are isolated from others in terms of broadcast.
- ✓ This **reduces network congestion**, improves **performance**, and makes **management easier**.

! Subnet Mask Tables

- **Class A, B, and C networks**
 - All can be **subnetted**.
 - Each class has a different number of bits for host/subnet info.
 - Subnetting rules depend on the network class used.
- **LAN subnetting**
 - Devices on the LAN understand subnetting details.
 - **External routers** need the **network part** of the IP.

- **Subnet sayısı** = $2^n \rightarrow (n = \text{eklenen subnet bit sayısı})$
- **Host sayısı/subnet** = $2^h - 2 \rightarrow (h = \text{kalan host bit sayısı})$

Table 10-5 IPv4 Class C subnet masks

Subnet mask	CIDR notation	Number of subnets on network	Number of hosts per subnet
255.255.255.128 or 11111111 11111111 11111111 10000000	/25	2	126
255.255.255.192 or 11111111 11111111 11111111 11000000	/26	4	62
255.255.255.224 or 11111111 11111111 11111111 11100000	/27	8	30
255.255.255.240 or 11111111 11111111 11111111 11110000	/28	16	14
255.255.255.248 or 11111111 11111111 11111111 11111000	/29	32	6
255.255.255.252 or 11111111 11111111 11111111 11111100	/30	64	2

! CIDR

Example: CIDR /25 :

- **IP adresinin ilk 25 biti network** kısmına aittir.
- **32 - 25 = 7 bit** ise **host** kısmına kalır.
- **Kalan host bit sayısı = 7** (çünkü $32 - 25 = 7$)
- Toplam kombinasyon: $2^7 = 128$
- Ama 2 tanesi **kullanılmaz**:
 - 1'i **network adresi**
 - 1'i **broadcast adresi**
- ◆ Bu yüzden: $128 - 2 = 126$ usable hosts

Class C adreslerinde ilk 24 bit network'tür (default).

- /25 dediğimizde: ekstra **1 bit daha** subnet'e verilmiş olur.
- Yani subnetler için **1 bit** kullanılmıştır: $2^1 = 2$ subnet

Class A: İlk 8 bit network için ayrılmıştır. CIDR: /8 Subnet Mask (dotted): 255.0.0.0

Class B: İlk 16 bit network için ayrılmıştır. CIDR: /16 Subnet Mask (dotted): 255.255.0.0

! Virtualization

Virtualization runs an **emulation** of a computer (with its own OS and apps) inside a real physical host.

- **Virtual machines (VMs)** include:
 - **Virtual workstations** (for desktop environments)
 - **Virtual servers** (for hosting services)
- You can assign different virtual hardware to each VM, such as:
 - **CPU cores**
 - **Storage drives**
 - **Network interface cards (NICs)**

Virtualization Concepts

- **VM Appearance**
 - A virtual machine (VM) looks and runs like a real computer to the user.
- **Host**
 - **The physical computer. it can** runs one or more VMs.
- **Guest**
 - **Each virtual machine** running on the host.
- **Hypervisor**
 - The software (also called a virtual machine manager) that configures, and controls VMs.

! Virtualization

• Advantages of virtualization

- Efficient use of resources
- Reduces cost and energy savings
- Fault and threat isolation , good security
- Simple backups, recovery and replication

• Disadvantages

- Can slow overall performance
- Increased complexity
- Increased licensing costs
- Creates single point of failure

Virtualization Software Examples

• VMware

- The most widely used virtualization platform.
- Offers tools for running multiple virtual workstations on one host.

• Other popular hypervisors:

- **Microsoft Hyper-V**
- **KVM (Kernel-based Virtual Machine)**
- **Oracle VirtualBox**
- **Citrix XenApp**

Each product provides similar core functions (creating and managing VMs) but differs in features, user interface, and ease of use.

Virtual Network Components

- A virtual network runs entirely inside one physical server, connecting VMs via **virtual network links** rather than physical hardware.
- Most networks combine physical and virtual elements

Virtual Machines and Network Adapters

- **Virtualization software** sets up each VM's virtual hardware and software through a user-friendly wizard.
- **Network connections** use **vNICs** (virtual network adapters):
 - Every VM needs at least one vNIC—and can have several.
 - When you create a vNIC, it's given a MAC address automatically.
 - By default, each vNIC is linked to a port on a virtual switch.

Virtual Switches and Bridges

- **When first VM's vNIC is selected**
 - The hypervisor makes a link between the VM's virtual NIC and the host
 - This link is called a bridge or virtual switch
- **Virtual Switch Features**
 - It's just software—no physical ports
 - Works at the Data Link layer
 - Connects VMs to each other
- **Hypervisor Management**
 - The hypervisor creates and controls all virtual switches
- **Traffic Flow**
 - VM packets go through their virtual switch to reach to network.

Network Connection Types

- You must choose a **networking mode** for each virtual NIC (vNIC).

- **Common network connections:**

- Bridged
- NAT
- Host-only

- **Bridged mode:**

- The vNIC uses the host machine's physical NIC to join the LAN.
- It gets its own IP address, default gateway, and netmask from DHCP server on physical LAN.
- It can connect with outer networks.
- It can be seen on real network.

- **NAT Mode**

- The vNIC uses the host as a NAT router.
- Obtains IP addressing information from the host. (host IP address = my own IP address).
- It cannot be seen on real network.
- The virtualization software acts as a DHCP server.
- Suitable for VMs that only need to make outgoing connections and don't need a fixed IP address.

- **Host-only Mode**

- VMs can exchange data with each other and with the host.
- They cannot reach or be reached by any external network. (outer world ile iletişim yok. sadece host ve diğer sanal makineler ile iletişimi var. Testler ve izole durumlar için kullanılır.)
- The host's physical NIC is not used for receive/transmit data.